

Software Management on Red Hat Based Systems



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Overview



Red Hat Software Tools

- Red Hat vs Debian
- Lab system
- Yum or Dnf
- RPMs and Modules
- Using yum/dnf
- Software repositories





Packaging Systems



Red Hat vs Debian

Red Hat

Uses RPM package files and the
YUM command

Debian

Uses DEB package files and the
APT command



Rocky Linux 9 Lab System



We are using a single Rocky 9 system

- VirtualBox / Vagrant
- vagrant up rocky



Demo



Review Lab Systems:

- List Vagrantfile
- Start and connect to Rocky Linux system





Red Hat Packaging Tools




```
$ ls -l /bin/dnf
lrwxrwxrwx. 1 root root 5 Jul 29  2020 /bin/dnf -> dnf-3
$ ls -l /bin/yum
lrwxrwxrwx. 1 root root 5 Jul 29  2020 /bin/yum -> dnf-3
```

Yum or Dnf?

Actually, it's neither. There is a lot of talk about the move to dnf, **DaNdiFied** Yum, in RHEL 8. But both dnf and yum are symbolic links to the command **dnf-3**. I use yum and will continue to use this symlink whilst it is available as it is most natural to use yum on RHEL systems




```
$ yum repolist
```

Content is Distributed via Two Main Repositories

A standard Red Hat 9 system with a support subscription will show two repositories.



Default Repositories

BaseOS

Provision of the core set of packages required for OS functionality. Packages are all in the form of RPM files

AppStream

This allows for additional user-space applications and services. Content is available in one of two formats: RPMs or the extension format called modules



Packaging

RPM Format

The Red Hat Package Manager of RPM format allows for applications in single standalone package files. The separation of the application or service from the required dependency packages allows for a single version of the application or service to be offered and supported

Module Format

The AppStream repository allows for modules. An extension to the RPM format grouping applications and services by version with their required dependencies. This extends the offering allowing for multiple versions of an application or service to be offered via the repository. Streams additionally have profiles allowing modular selection of what should be installed



Demo



For our first demonstration we will:

- List available repositories
- Check the version of the dnf-3 command





Using YUM




```
$ yum list tree*
```

```
$ sudo yum install tree
```

```
$ sudo yum remove -y tree
```

```
$ sudo yum history list
```

```
$ sudo yum history undo 14
```

```
$ sudo yum update
```

```
$ sudo yum history undo 16
```

◀ Check if package is installed or not, globbing can be used

◀ Install software package, use **-y** to install without interaction

◀ Removing an installed package, **purge** would also remove configuration files

◀ View the yum/dnf history log

◀ Undo the numbered command in your history. This can be quite useful in reversing an update.

◀ Update all packages, this will install new Kernel packages if available. If this upsets the system we can reverse the update



Demo



You will now start using yum:

- Search and list packages
- Install packages
- Remove packages
- Replay yum history





Managing Software Repositories





Software Repositories

The big advantage package managers offer is the use of online repositories providing access to the software. We can also create our own repositories in the local filesystem or through a web server



```
# ls /etc/yum.repos.d  
# yum repolist  
# yum repolist --all
```

Working with Repos

Repository files are located in the `/etc/yum.repos.d/` directory and use the extension `.repo`. A single repo file may contain more than one repository source.



```
$ sudo rpm -qp1 https://yum.puppet.com/puppet7-release-el-9.noarch.rpm
$ sudo dnf install https://yum.puppet.com/puppet7-release-el-9.noarch.rpm
$ cat /etc/yum.repos.d/puppet7-release.repo
$ yum repolist
$ sudo dnf install -y puppet-agent
```

Add 3rd Part Repo

To install Puppet Configuration Management tools, ProgressChef provide an RPM to add the public key and repository file. This can be installed using the RPM command



Demo



Manage Repositories:

- List repositories
- Add repository





Security Updates



```
$ yum updateinfo
$ yum updateinfo --security
$ yum updateinfo --sec-severity Important
$ yum updateinfo info --sec-severity Important
$ sudo yum update --sec-severity Important
```

Update Metadata

Metadata is maintained for all packages including for updates. We can choose to filter on security updates, installing only the severity that we need.



Demo



Reading metadata from updates

- List updates and their level
- Install updates with a higher security implication



Summary



Software management in Red Hat Based systems is provided by yum/dnf

yum/dnf

- install
- erase
- list
- provides

repos

- /etc/yum.repos.d/*.repo
- yum repolist

Up Next:

Managing Security Using SELinux

